

Fintech Review Analytics - Interim Submission Report

Due date from PDF: 18 May 2026, 8:00 PM UTC (interim submission).

Scope

- Task 1 is complete on main: scraping, preprocessing, .gitignore, CI, requirements, tests, and documentation are present.
- Task 2 partial progress is complete on main: VADER sentiment labels/scores, identified themes, and TF-IDF keyword evidence are implemented.
- The repository history includes the merged task-2 branch.

Scraping and Data Quality

- Google Play scraper collected 500 reviews each for Commercial Bank of Ethiopia, Bank of Abyssinia, and Dashen Bank, for 1,500 raw reviews total.
- Preprocessing removed duplicates by review_id, dropped missing review_text/rating rows, stripped blank text, and normalized dates to YYYY-MM-DD.
- Current clean output contains 1,500 rows; missing review_text = 0 and missing rating = 0 after cleaning.
- Data files are intentionally ignored by git and can be reproduced with the README commands.

Early Sentiment Findings

- Bank of Abyssinia: 500 reviews, average rating 3.57, average VADER sentiment 0.172, negative share 17.4%.
- Commercial Bank of Ethiopia: 500 reviews, average rating 4.11, average VADER sentiment 0.274, negative share 10.6%.
- Dashen Bank: 500 reviews, average rating 3.93, average VADER sentiment 0.290, negative share 12.8%.

Thematic Progress

- Keyword rules currently assign draft themes: Account Access, Transaction Performance, Reliability and Crashes, UI and Usability, Customer Support, and General Feedback.
- TF-IDF output is generated by bank and sentiment label to support final grouping into 3-5 themes per bank.

Blockers and Final Plan

- Google Play review availability can vary by region, language, and app, so fresh reruns may produce a different date window.
- VADER is used for interim speed and reproducibility; DistilBERT comparison can be added for the final report if time allows.
- Draft theme labels are keyword-based and need manual review against TF-IDF output before final recommendations.
- Next: finalize theme grouping, add PostgreSQL schema/loading workflow, and produce stakeholder-ready recommendations and final PDF report.

Required Interim Visualization

